

ANNA GEORGE

Computer Science & Engineering Final Year Student — Specialized Honors in Machine Learning

Kerala, India

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Languages: English (Fluent), French (Moderate), Malayalam (Native), Hindi (Fluent)

Education

Mar Baselios College of Engineering & Technology

Thiruvananthapuram, India

Bachelor of Technology in Computer Science and Engineering (Final Year Student)

Specialization: Specialized Honors in Machine Learning

Academic Standing: **Institutional Topper** (Highest Cumulative CGPA: **9.85 / 10.00**)

International Mobility: Chosen as **1 of only 5 students institution-wide** for an international academic exchange program at **Universiti Teknologi Malaysia (UTM)**, validating cross-cultural adaptability.

Experience

Genrobotic Innovations Private Limited

Technopark, India

Research & Development Intern

12/2024 – 01/2025

- Engineered and tested autonomous navigation algorithms in **Python** using **ROS2**, utilizing **Gazebo** and **RViz2** to simulate, debug, and optimize real time sensor data processing.
- Programmed a unified code base enabling seamless transition from digital twin simulations to physical hardware deployment.

Intel Unnati Labs

Thiruvananthapuram, India

Industrial Trainee

04/2025 – 06/2025

- Selected via **competitive institutional ranking** to deploy high performance Machine Learning frameworks. Operated within an industry aligned laboratory testing environment.

Notable Projects

Adaptive Cyber Deception Platform with Applied AI

06/2026 – Present

Applied AI Infrastructure

- Designing and aiming to deploy a proactive cyber security defense network utilizing advanced Retrieval-Augmented Generation (**RAG**) architectures and Large Language Models (**LLMs**).
- Integrating intelligent, dynamic streaming data queries to orchestrate deception tactics, safeguarding complex network flows against advanced adversarial interference.

Multi Model AQI Time Series Forecasting Framework

11/2025 – 12/2025

Predictive Modeling & Advanced Data Science

- Formulated a comparative time-series analytical layout utilizing national daily Indian AQI data streams to transition environmental public policy from static regression to dynamic, temporal modeling.
- Architected, trained, and calibrated three advanced predictive setups: a statistical standard (**SARIMA**), a lag-engineered machine learning framework (**XGBoost**), and a deep sequential neural network (**LSTM**).
- Implemented a rigorous 30-day blind-test split matrix and evaluated variance patterns using **MAE**, **RMSE**, and **MAPE** metrics to isolate forecasting anomalies and guarantee extreme baseline integrity.

Agile Blockchain Based Clinical Data Integrity Network

01/2026 – 05/2026

Secure Architecture & Decentralized Governance

- Created a decentralized, tamper-proof Clinical Trial Management System utilizing **Hyperledger**, **React**, **MongoDB**, and **Docker** to dramatically scale data legitimacy, transparency, and clinical trial safety.
- Replaced manual, rule based logging with cryptographic **SHA-256 hashing** to secure sensitive metadata, preventing variations and structural corruption across multi party transaction schemes.
- Managed the product life cycle within a strict **Agile framework**, driving sprint level updates, iterative peer code reviews, and demonstrable, production oriented milestone outcomes.

KMRL Unstructured Data Extraction & Processing System

04/2025

Smart India Hackathon (SIH) – 12-Hour Rapid Prototyping Pipeline

- Fabricated a text parsing solution to isolate critical, risk related data from multi format unstructured files (PDFs, documents, and images).
- Directed an intelligent data analysis workflow integrating the **Groq LLM API** to scan source documents, analyze semantic anomalies, and filter predefined emergency keywords.

- Integrated an Optical Character Recognition (**OCR**) vision layer to ingest text data from image files, automatically transforming raw telemetry inputs into structured, actionable **JSON schemas**.

Leadership & Engagement

- **Primary Conference Organizer:** Structured and coordinated technical operations for the **3rd International Conference on Networks and Advancements in Computational Technologies**, managing multi disciplinary agendas and logistical timelines.
- **Event Coordinator:** Supervised logistical operations and collaborative **sprint planning** for the Indian Society for Technical Education (**ISTE**) Kerala Chapter regional technology event.

Technical Skills

- **Programming Languages:** Python, Java, C, SQL (Postgres), HTML/CSS
- **Applied AI & Data Analytics:** RAG Pipelines, LLMs, Time-Series Forecasting (SARIMA/LSTM), Anomaly Detection Architectures, Statistical Process Control, Hyperledger Blockchain, Cryptographic Hashing (SHA-256)
- **Frameworks & Libraries:** Pandas, NumPy, Matplotlib, Seaborn, ROS2
- **Developer Tools & Ecosystems:** Git, Docker, Gazebo, RViz2, Linux